Task-4

(1) Ans: First, I will use the given GPS coordinates of the grid's origin as the reference point. Then, I will convert the rover's GPS latitude and longitude into local East-north coordinates. Next, I will determine the grid's rotation angle using the known grid orientation or reference landmarks. After that, I will rotate the local coordinates into the grid coordinate system to identify the rover's correct position in the 2×2 grid. The rover's own heading is not required because GPS-based position and grid orientation can be used to determine its location.

Justification: The grid is not perfectly aligned with the compass directions. Therefore, a rotation transformation is necessary to calculate the correct grid position

(2) Ans: The problem may be caused by both GPS and IMU nodes using the same USB device path or serial port configuration. When the IMU starts, it may take control of the port that was already being used by the GPS. After rebooting, the device connection order may change, which can cause the problem to switch between the two sensors

Solution:

(1) Check the USB device paths and serial port configurations.
(2) Assign a unique port to each sensor.
(3) Use stable device names or unique port device IDs instead of temporary port names.
(4) Make sure both nodes have separate configurations.
(5) Test both sensors together and check that they still work after rebooting.

Justification:

Using unique and stable device identifiers prevents both sensors from accessing the same port. It also ensures reliable communication after restarting the rover's computer.